Experiment Report - regression

14 configurations of 3 model(s), ranked by r2 (higher is better) · 2026-11-11 22:39

1. Executive Summary

best configuration	Ridge (rank #1)
r2	0.4983 (higher is better)
all metrics	r2 0.4983, mae 44.2742, mse 2975.2465, rmse 54.5458
hyper-parameters	{"alpha": 0.01}
features	10: age, sex, bmi, bp, s1, s2, s3, s4, s5, s6
validation	k_folds
configurations / models	14 / 3
visualizations	17 generated, 2 skipped, 0 failed

- Ridge leads Lasso by 0.0004 r2 (best configuration of each).
- Lasso varies most across its configurations (std of r2 0.2029).
- Training time per configuration: median 0.008 s, total 5.6 s of worker time.
- The best of many configurations is optimistically biased; confirm on held-out data.

2. Experiment Configuration

created	2026-11-11 22:39:31
task	regression
validation	k_folds
ranking metric	r2 (higher is better)
models	Ridge, Lasso, RandomForestRegressor
preprocessing	none
feature subsets tried	1
grid: alpha	[0.01, 0.1, 1, 10]
grid: max_depth	[3, 6, null]
grid: min_samples_leaf	[1, 10]

3. Dataset Overview

rows	442
features	10

Feature columns: age, sex, bmi, bp, s1, s2, s3, s4, s5, s6

4. Overall Results and Model Comparison

Per model

Best, mean, spread and worst r2 over each model's configurations, and the mean of every other metric.

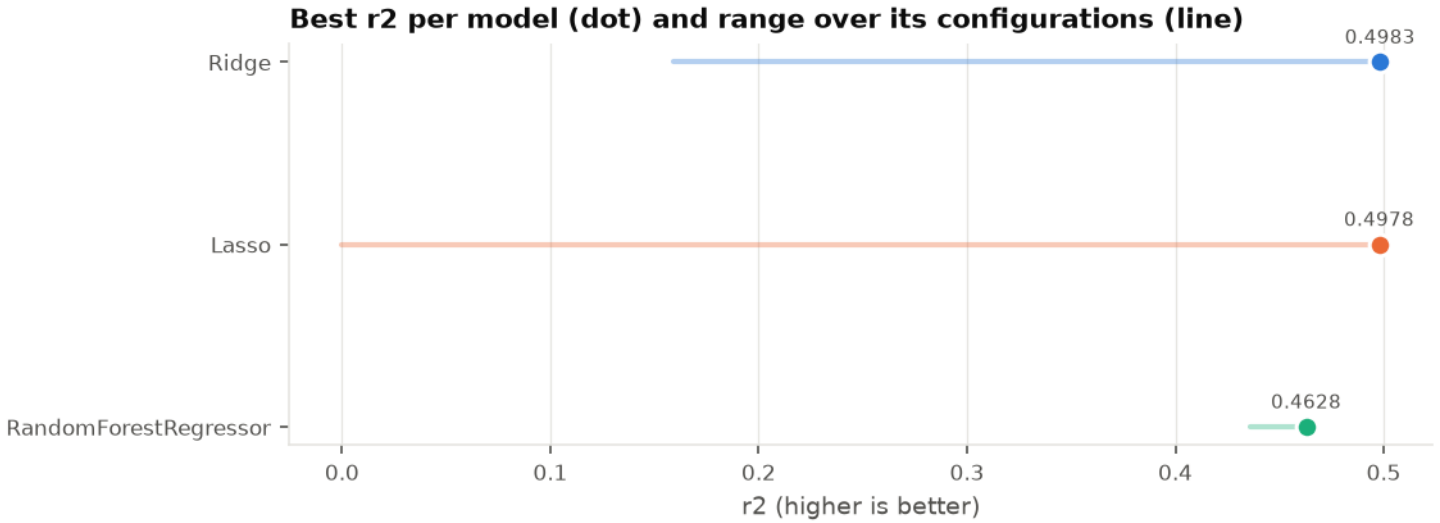
model	configurations	best_r2	mean_r2	std_r2	worst_r2	mean_mae	mean_mse	mean_rmse	best_rank	best_n_features
Ridge	4	0.4983	0.3949	0.1391	0.1593	49.5102	3588.1183	59.5393	1	10
Lasso	4	0.4978	0.3350	0.2029	-0.0003	51.9260	3943.4388	62.1294	2	10
RandomForestRegressor	6	0.4628	0.4510	0.0095	0.4357	46.4106	3255.3436	57.0535	5	10

Top 14 configurations

rank	model	params	n_features	validation	r2	mae	mse	rmse
1	Ridge	{"alpha": 0.01}	10	k_folds	0.4983	44.2742	2975.2465	54.5458
2	Lasso	{"alpha": 0.01}	10	k_folds	0.4978	44.2984	2977.8384	54.5696
3	Ridge	{"alpha": 0.1}	10	k_folds	0.4965	44.4870	2985.6038	54.6407
4	Lasso	{"alpha": 0.1}	10	k_folds	0.4948	44.6447	2995.4934	54.7311
5	RandomForestRegressor	{"max_depth": null, "min_samples_leaf": 10}	10	k_folds	0.4628	45.8353	3185.5880	56.4410
6	RandomForestRegressor	{"max_depth": 6, "min_samples_leaf": 10}	10	k_folds	0.4619	45.8711	3191.1580	56.4903
7	RandomForestRegressor	{"max_depth": 3, "min_samples_leaf": 10}	10	k_folds	0.4514	46.5442	3253.0653	57.0356
8	RandomForestRegressor	{"max_depth": 3, "min_samples_leaf": 1}	10	k_folds	0.4503	46.6027	3259.8585	57.0952
9	RandomForestRegressor	{"max_depth": 6, "min_samples_leaf": 1}	10	k_folds	0.4441	46.7017	3296.3998	57.4143
10	RandomForestRegressor	{"max_depth": null, "min_samples_leaf": 1}	10	k_folds	0.4357	46.9088	3345.9920	57.8446
11	Ridge	{"alpha": 1}	10	k_folds	0.4255	48.8406	3406.4356	58.3647
12	Lasso	{"alpha": 1}	10	k_folds	0.3475	52.9942	3868.9535	62.2009
13	Ridge	{"alpha": 10}	10	k_folds	0.1593	60.4392	4985.1873	70.6059
14	Lasso	{"alpha": 10}	10	k_folds	-0.0003	65.7669	5931.4697	77.0160

Model ranking

id: model_ranking · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)

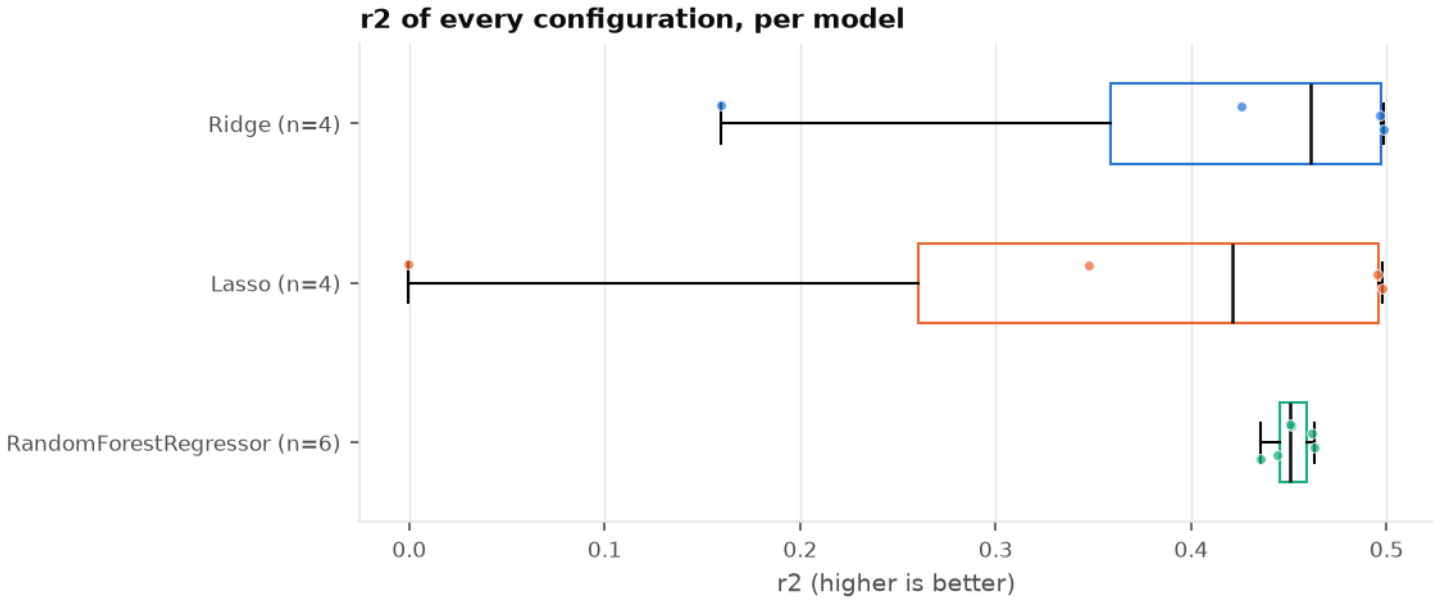


What it shows. Each model's best configuration (dot) and the range of the ranking metric over all of its configurations (line), best model on top. A long line means the model is sensitive to its hyper-parameters / feature subset. The best of many tries is optimistically biased: compare with the spread, not only the dot.

Ridge best r2	0.4983
Ridge configurations	4
Lasso best r2	0.4978
Lasso configurations	4
RandomForestRegressor best r2	0.4628
RandomForestRegressor configurations	6

Score distribution across configurations

id: config_distribution · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)

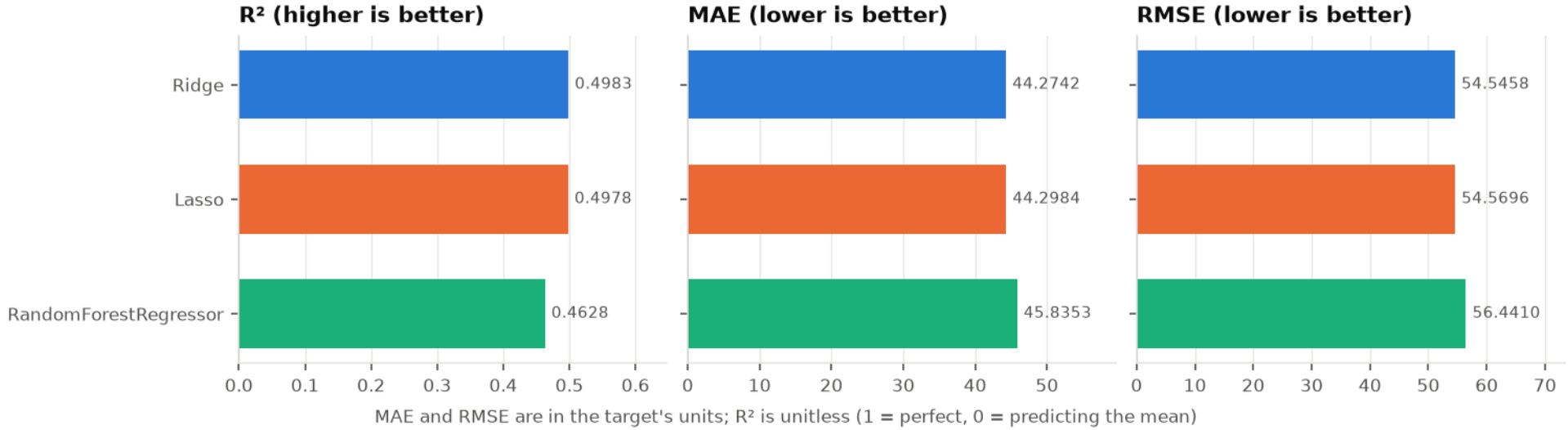


What it shows. Box plot of the ranking metric over every configuration (hyper-parameters x feature subset) of each model, with every configuration drawn as a point. Shows how robust a model family is to its settings - a narrow box near the top is a safer choice than a single lucky configuration.

Ridge median r2	0.4610
Lasso median r2	0.4212
RandomForestRegressor median r2	0.4508

Regression metrics by model

id: reg_metrics · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)



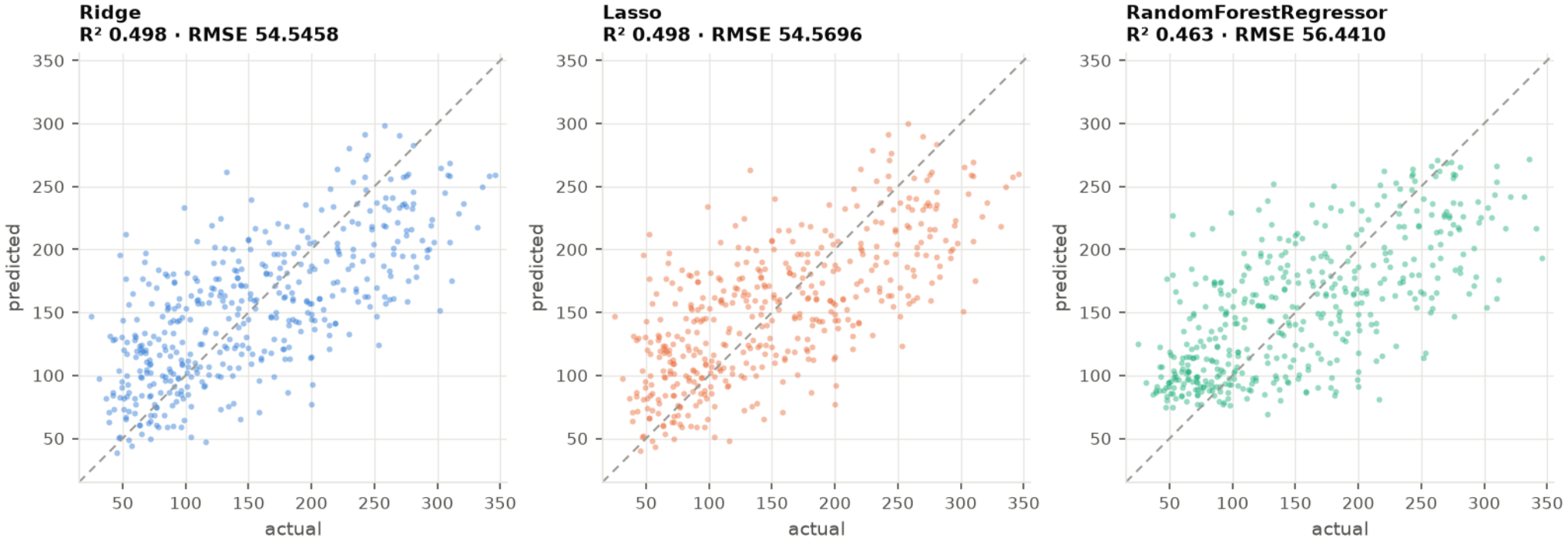
What it shows. R², MAE and RMSE of each model's best configuration, each on its own axis because their scales differ. RMSE weighs large errors more than MAE: a model with a low MAE but a high RMSE makes occasional big mistakes. (MSE = RMSE² is listed below.)

Ridge	r2 0.4983, mae 44.2742, mse 2975.2465, rmse 54.5458
Lasso	r2 0.4978, mae 44.2984, mse 2977.8384, rmse 54.5696
RandomForestRegressor	r2 0.4628, mae 45.8353, mse 3185.5880, rmse 56.4410

5. Regression Analysis

Actual vs predicted

id: reg_actual_vs_predicted · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)

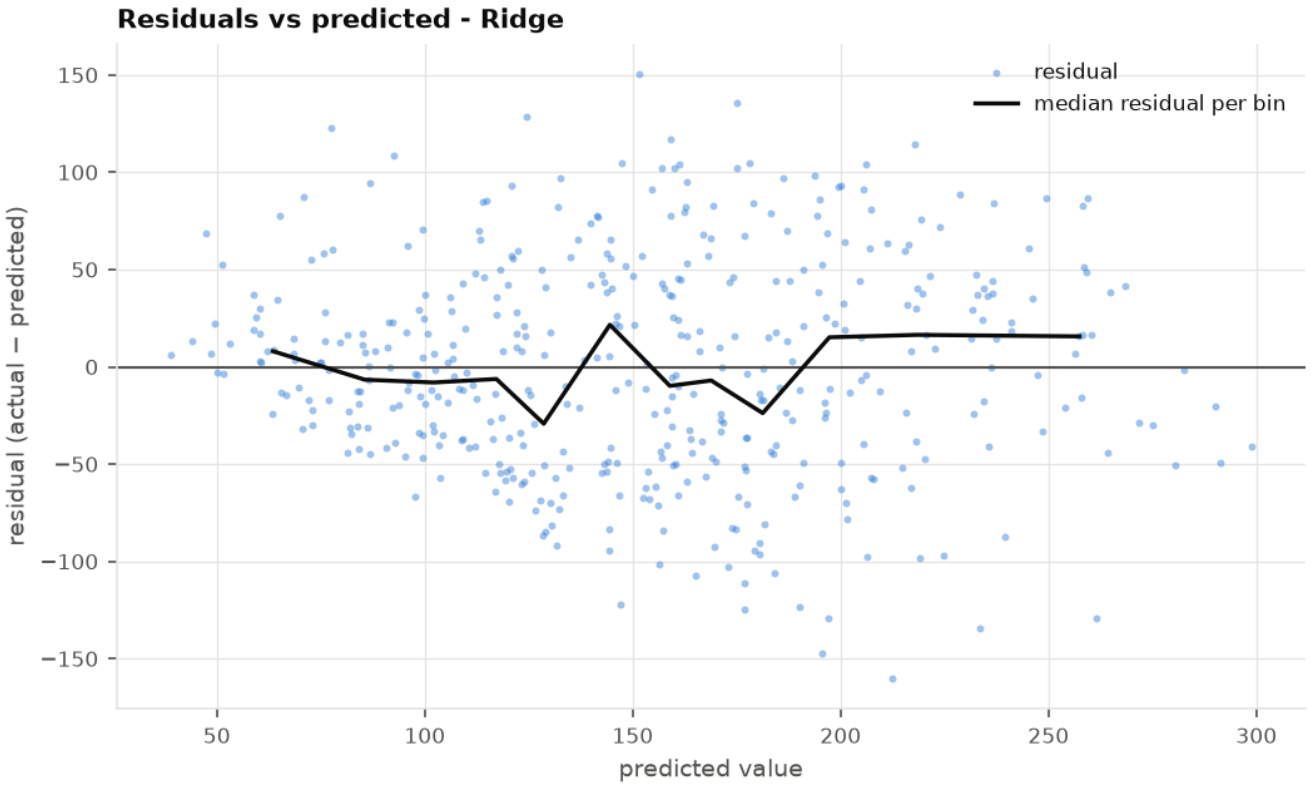


What it shows. Each out-of-sample prediction against its true value, one panel per model on identical axes. Points on the dashed identity line are perfect; a cloud flatter than the line shows the model shrinks toward the mean, and a widening cloud shows errors that grow with the target.

Ridge	R² 0.4983, RMSE 54.5458
Lasso	R² 0.4978, RMSE 54.5696
RandomForestRegressor	R² 0.4628, RMSE 56.4410

Residuals vs predicted values

id: reg_residuals · model(s): Ridge (rank #1)

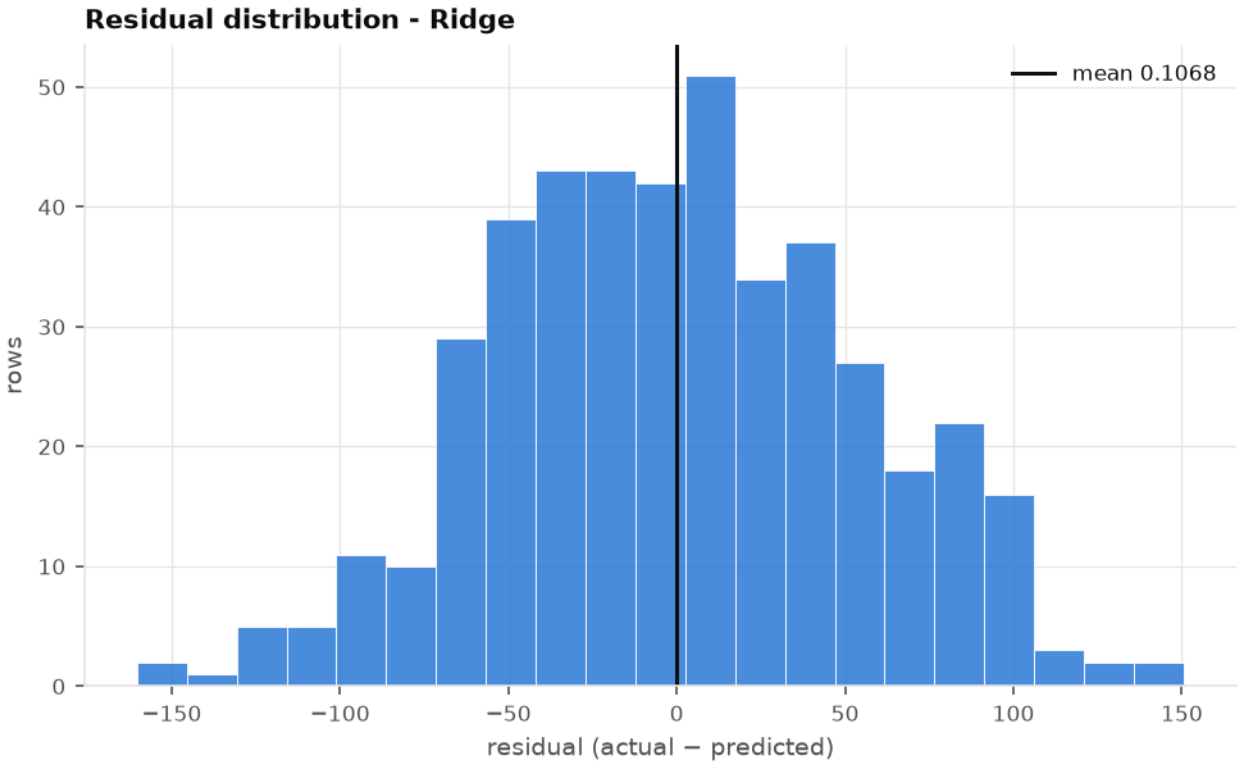


What it shows. Residuals of the best model against its predictions, with the median residual per prediction bin. Ideally a flat, even band around zero. A curved median line means a missed non-linearity; a funnel shape means the error grows with the prediction (heteroscedasticity).

mean residual (bias)	0.1068
residual SD, lower / upper half of predictions	47.0351 / 61.1301

Residual distribution

id: reg_residual_distribution · model(s): Ridge (rank #1)

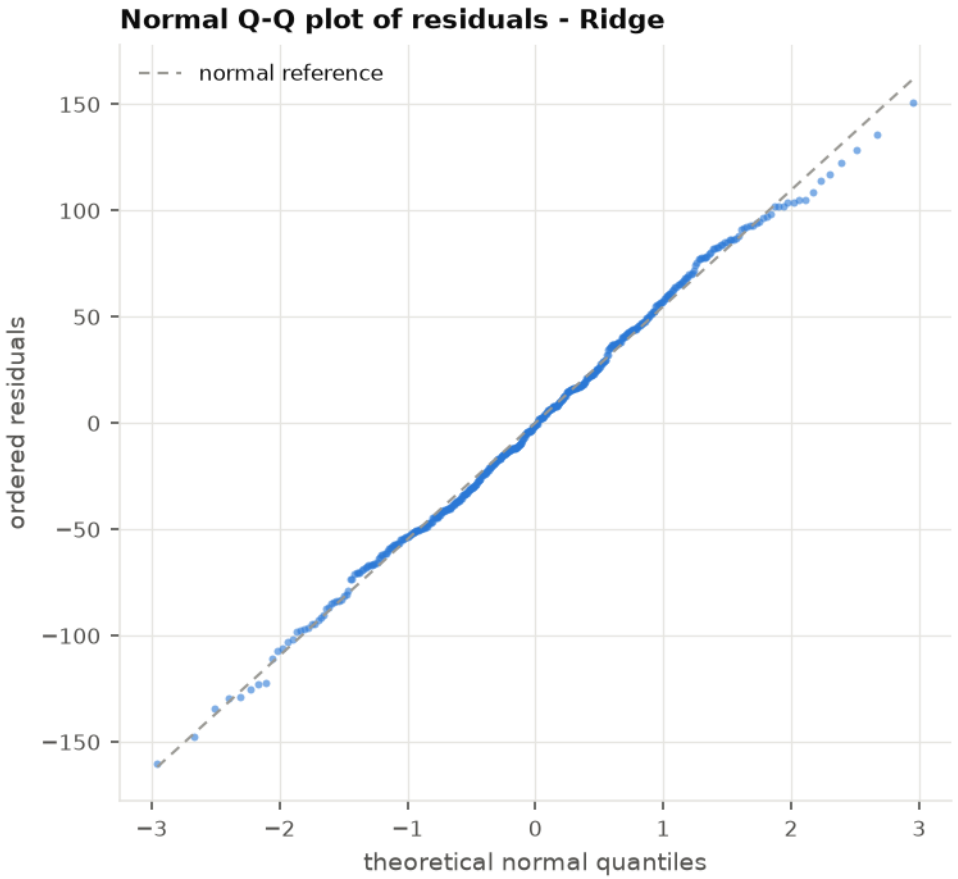


What it shows. Histogram of the best model's residuals. Centred on zero means no overall bias; skew or a long tail shows the model systematically under- or over-predicts some rows.

mean	0.1068
standard deviation	54.6075
skewness	0.0037
rows	442

Normal Q-Q plot of residuals

id: reg_qq · model(s): Ridge (rank #1)

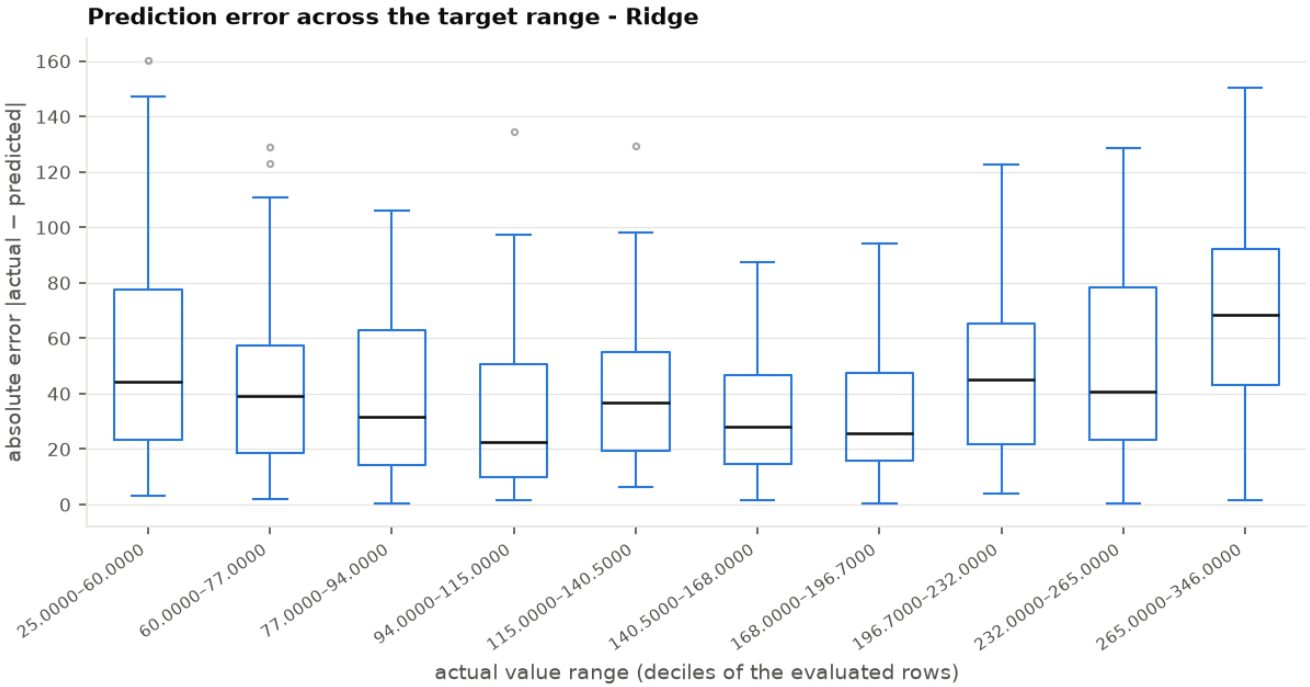


What it shows. Residual quantiles against those of a normal distribution. Points on the line = normally distributed errors; ends bending away = heavier tails than normal, so intervals that assume normal errors would be too narrow. A diagnostic, not a requirement of the models.

probability-plot correlation r	0.9986

Error by target range

id: reg_error_by_range · model(s): Ridge (rank #1)

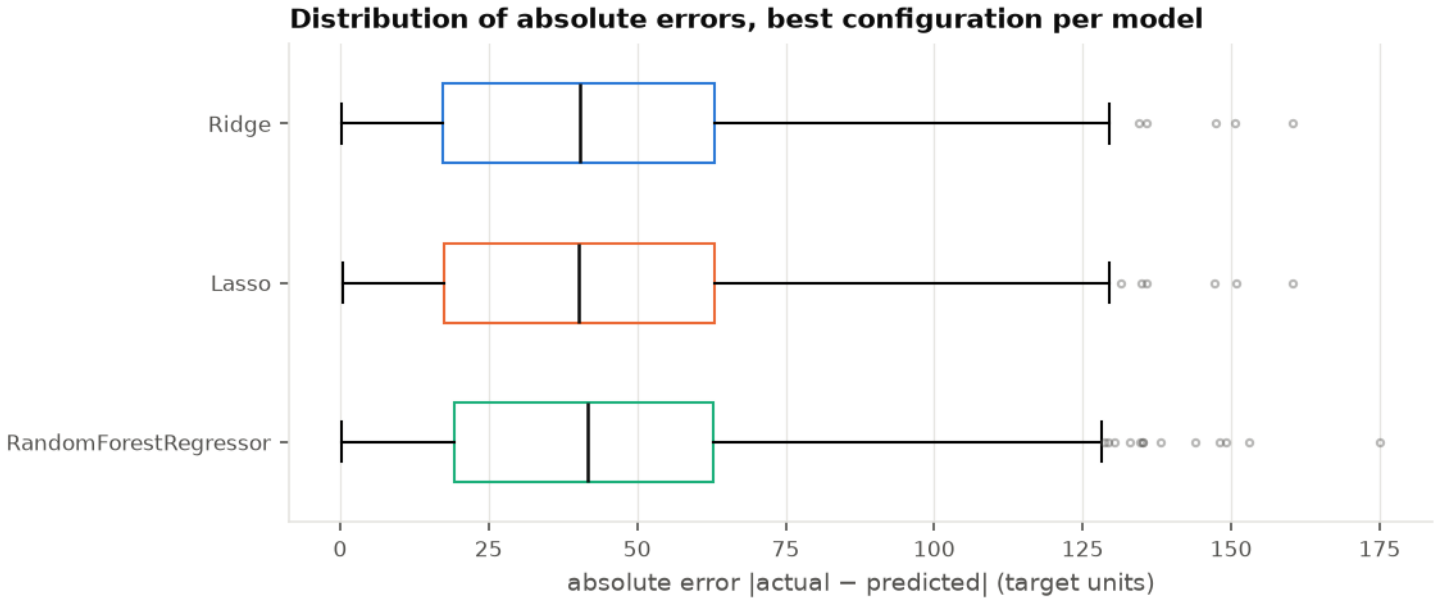


What it shows. Absolute errors of the best model grouped by deciles of the true value. Shows whether the model is reliable across the whole range or fails at the extremes (often where training data is scarce).

MAE	44.2742
worst range (median |error|)	265.0000–346.0000 (68.3922)

Error distribution across models

id: reg_error_models · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)



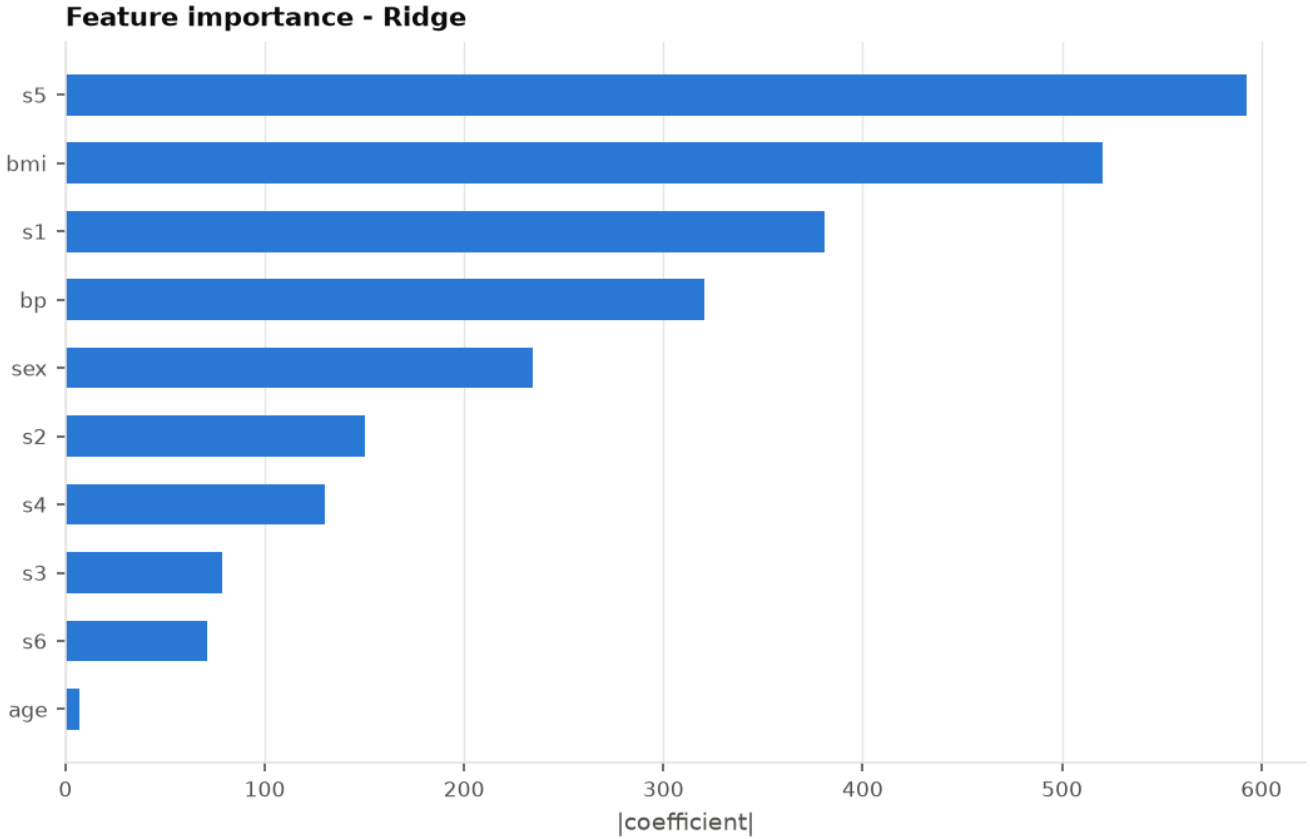
What it shows. Box plots of every out-of-sample absolute error for each model's best configuration. Two models with the same MAE can differ in their tails: the 90th percentile and the outliers show how bad the worst predictions get.

Ridge	median 40.3671, 90th percentile 91.2240
Lasso	median 40.2220, 90th percentile 91.1061
RandomForestRegressor	median 41.6303, 90th percentile 90.6316

6. Feature Analysis

Feature importance - Ridge

id: feature_importance:Ridge · model(s): Ridge (rank #1)



What it shows. The importance the fitted model itself assigns to each input: feature_importances_ for tree ensembles, the absolute coefficient for linear models. Shows what the model relies on, not causal effects; correlated features share importance. For a model-agnostic, held-out measure use FeatureResearch.permutation_importance().

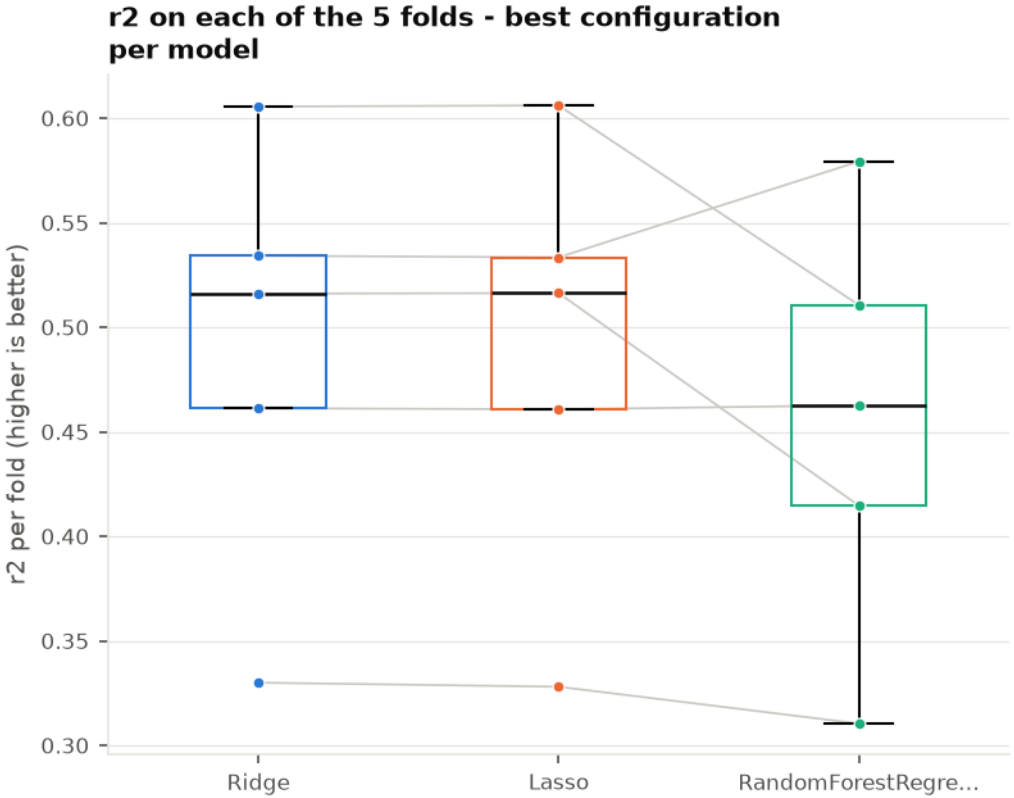
Inputs were not standardized: coefficient sizes depend on each feature's units.

s5	592.3480
bmi	520.5886
s1	380.6071
bp	320.5171
sex	234.5498
s2	150.4847
s4	130.3125
s3	78.5893
s6	71.1348
age	7.1975

7. Cross-Validation Analysis

Metric distribution across cross-validation folds

id: cv_folds · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)



What it shows. The ranking metric computed separately on each validation fold for each model's best configuration. If the gap between two models is small compared with the fold-to-fold spread, the ranking is not reliable. Folds share training rows, so the spread understates the true variance and no significance test is implied.

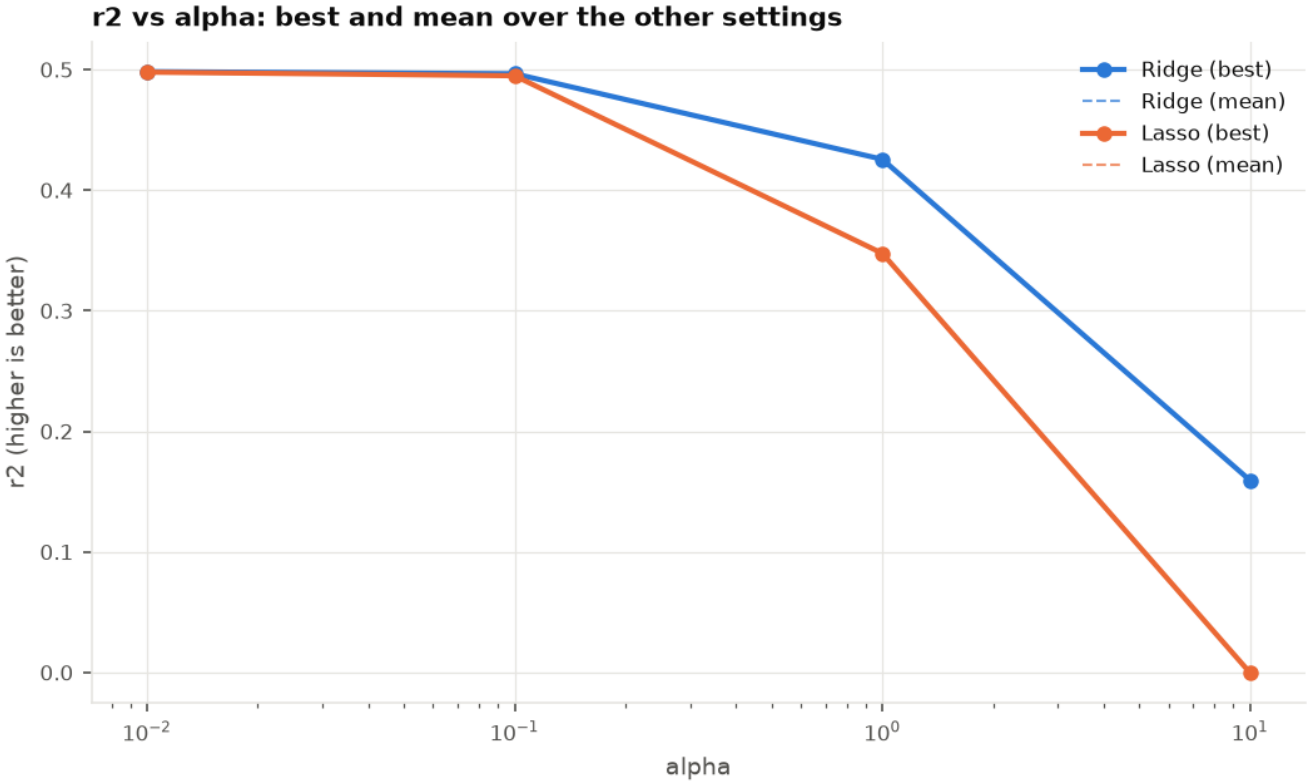
Grey lines join the same fold across models (every model is scored on identical folds).

Ridge r2 mean ± sd	0.4894 ± 0.1030
Lasso r2 mean ± sd	0.4889 ± 0.1038
RandomForestRegressor r2 mean ± sd	0.4555 ± 0.1014

8. Hyperparameter Optimization

Hyper-parameter sensitivity - alpha

id: hp_sensitivity:alpha · model(s): Ridge (rank #1), Lasso (rank #2)

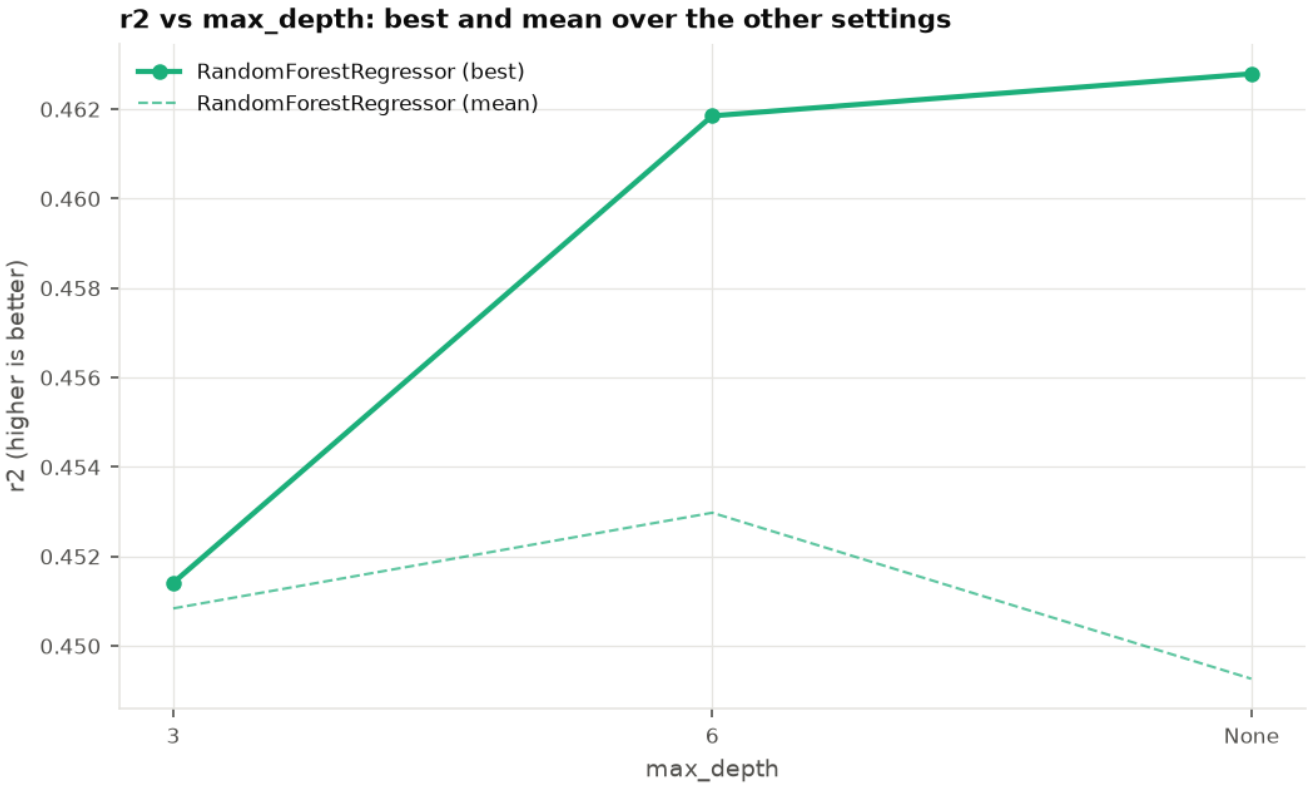


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

Ridge best alpha	0.01 (r2 0.4983)
Lasso best alpha	0.01 (r2 0.4978)

Hyper-parameter sensitivity - max_depth

id: hp_sensitivity:max_depth · model(s): RandomForestRegressor (rank #5)

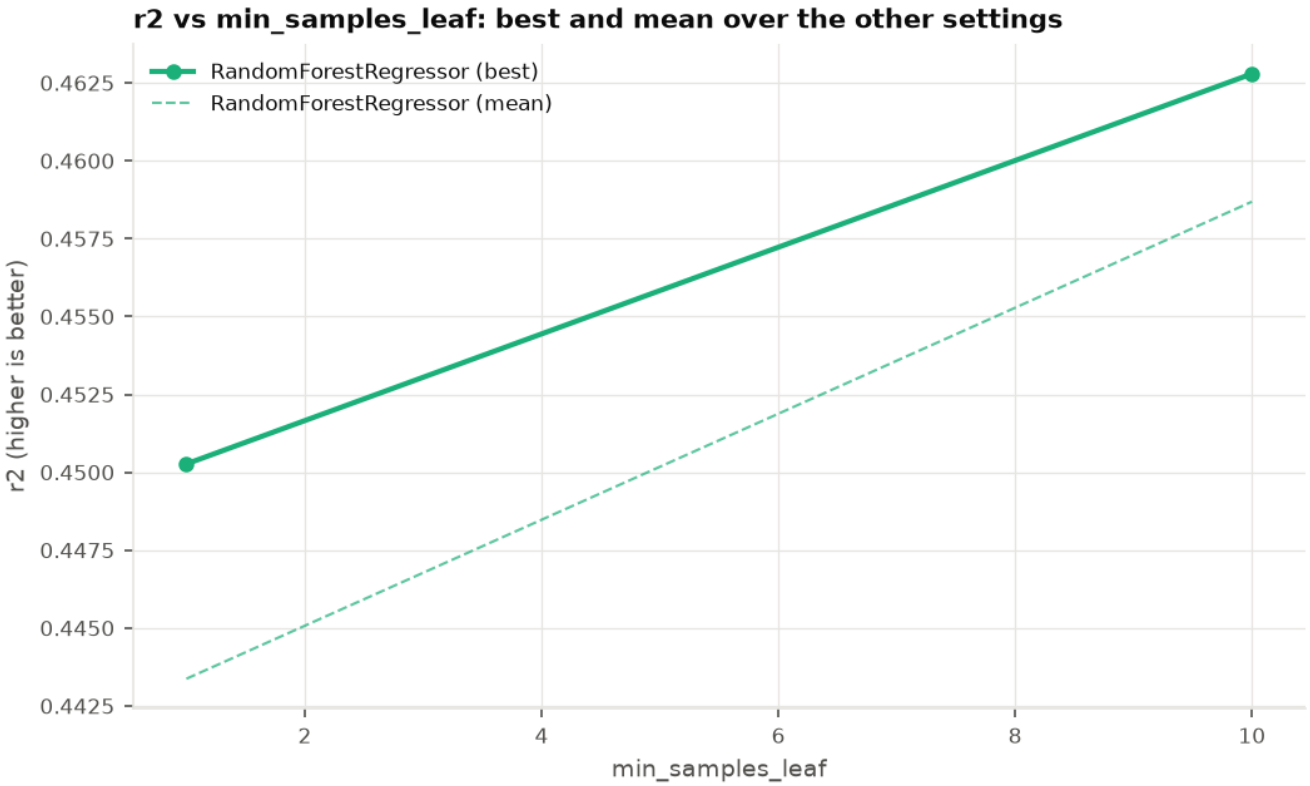


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

RandomForestRegressor best max_depth	None (r2 0.4628)

Hyper-parameter sensitivity - min_samples_leaf

id: hp_sensitivity:min_samples_leaf · model(s): RandomForestRegressor (rank #5)

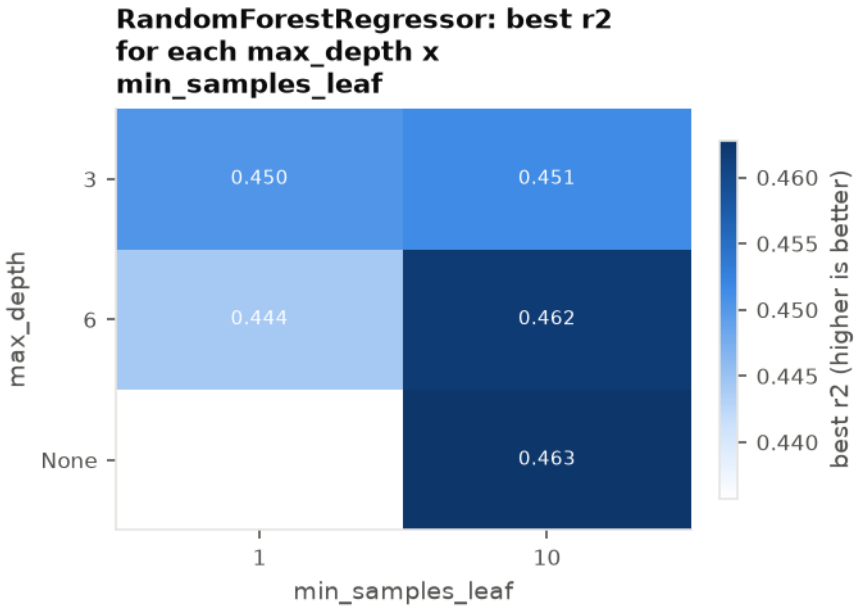


What it shows. A validation curve built from the grid already trained: for each value of one hyper-parameter, the best (solid) and mean (dashed) ranking metric over every other setting, per model. A flat curve means the parameter barely matters; a best value at the edge of the grid suggests extending the grid. Marginal view - interactions are in the heatmap.

RandomForestRegressor best min_samples_leaf	10 (r2 0.4628)

Two-parameter interaction - RandomForestRegressor

id: hp_heatmap:RandomForestRegressor · model(s): RandomForestRegressor (rank #5)



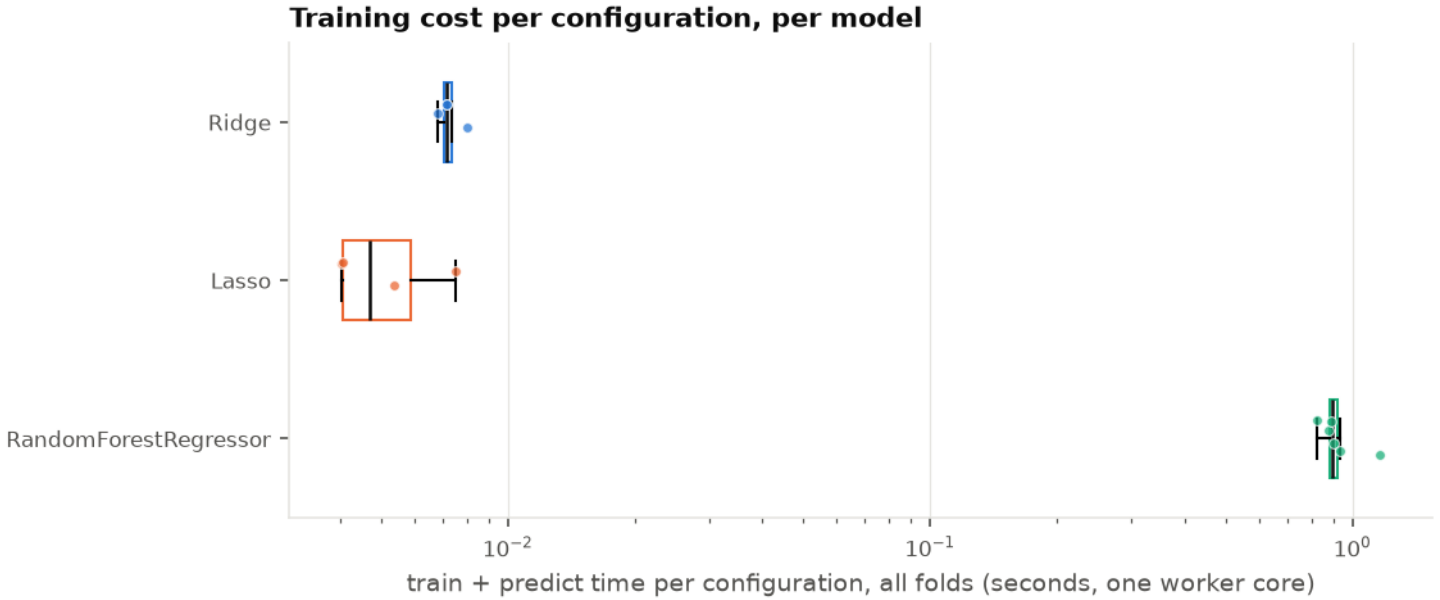
What it shows. Best ranking metric for every combination of the model's two most-varied hyper-parameters (the best over any other settings in each cell). Diagonal ridges reveal interactions that the one-parameter curves cannot show. Darker = larger value; check the colour-bar direction note.

best cell	max_depth=None, min_samples_leaf=10 (r2 0.4628)

9. Runtime and Resource Analysis

Runtime comparison

id: runtime · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)

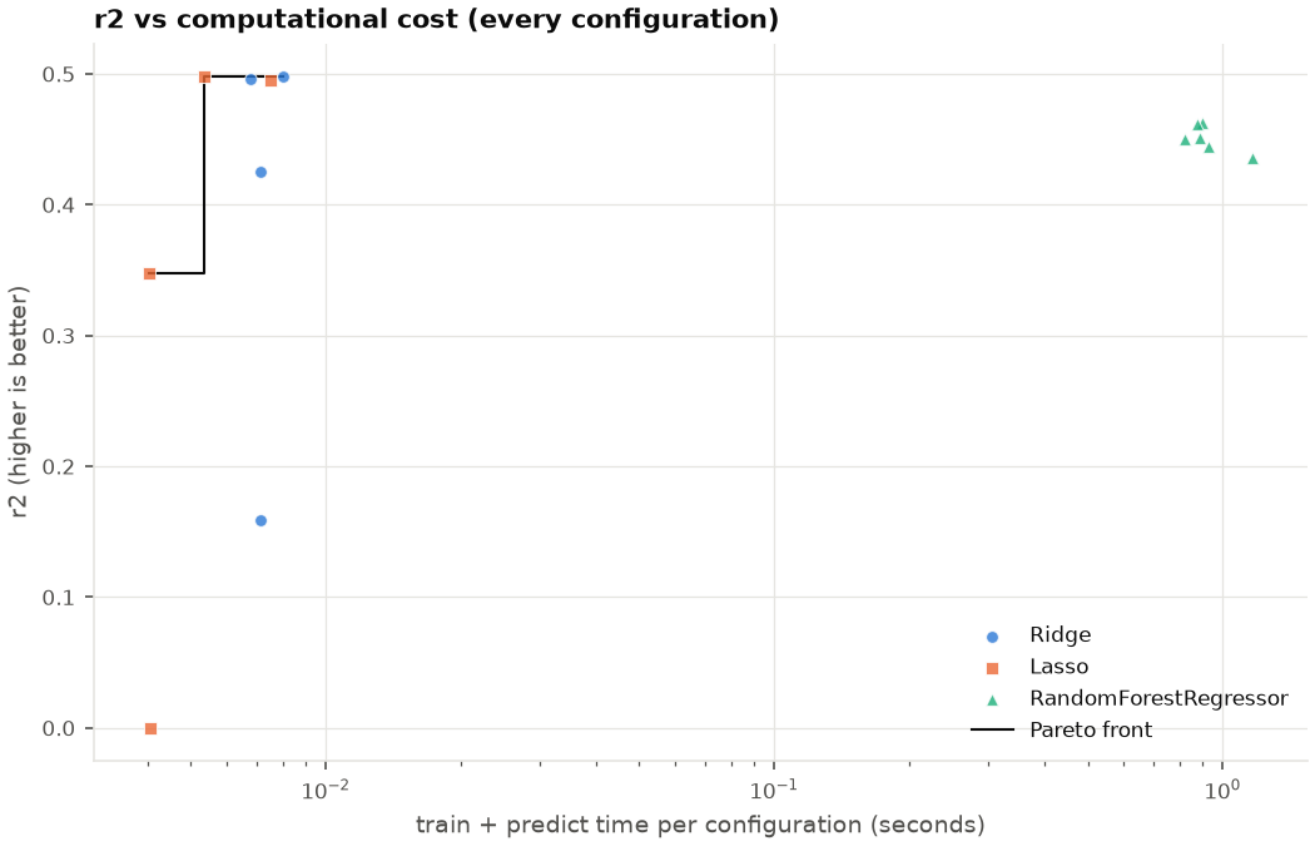


What it shows. Wall-clock time of each configuration (fit and predict over every fold), measured inside the worker process on one core. Includes preprocessing inside the pipeline; excludes data transfer. Absolute values depend on the machine and its load; compare models within one run.

Ridge median / total	0.007 s / 0.03 s
Lasso median / total	0.005 s / 0.02 s
RandomForestRegressor median / total	0.896 s / 5.59 s

Performance vs computational cost

id: cost_vs_score · model(s): Ridge (rank #1), Lasso (rank #2), RandomForestRegressor (rank #5)



What it shows. Every configuration's score against its training time. The Pareto front joins the configurations that no other one beats on both score and time: a configuration near the top of the front but much further left is almost as good for a fraction of the cost.

configurations on the Pareto front	3
fastest configuration	0.004 s (r2 0.3475)
best configuration	0.008 s (r2 0.4983)

10. Visualization Index

Every visualization attempted for this report: 17 generated, 2 skipped, 0 failed.

#	visualization	section	status	time (s)	detail
1	Model ranking	Overall Results and Model Comparison	SUCCESS	0.10	included in the report
2	Score distribution across configurations	Overall Results and Model Comparison	SUCCESS	0.11	included in the report
3	Metric distribution across cross-validation folds	Cross-Validation Analysis	SUCCESS	0.13	included in the report
4	Hyper-parameter sensitivity - alpha	Hyperparameter Optimization	SUCCESS	0.21	included in the report
5	Hyper-parameter sensitivity - max_depth	Hyperparameter Optimization	SUCCESS	0.11	included in the report
6	Hyper-parameter sensitivity - min_samples_leaf	Hyperparameter Optimization	SUCCESS	0.13	included in the report
7	Two-parameter interaction - RandomForestRegressor	Hyperparameter Optimization	SUCCESS	0.13	included in the report
8	Feature selection: score vs number of features	Feature Analysis	SKIPPED	0.00	every configuration used the same number of features
9	Runtime comparison	Runtime and Resource Analysis	SUCCESS	0.15	included in the report
10	Performance vs computational cost	Runtime and Resource Analysis	SUCCESS	0.17	included in the report
11	Memory consumption comparison	Runtime and Resource Analysis	SKIPPED	0.00	memory consumption is not measured per configuration by the pipeline; no values are estimated
12	Regression metrics by model	Overall Results and Model Comparison	SUCCESS	0.16	included in the report
13	Actual vs predicted	Regression Analysis	SUCCESS	0.33	included in the report
14	Residuals vs predicted values	Regression Analysis	SUCCESS	0.19	included in the report
15	Residual distribution	Regression Analysis	SUCCESS	0.10	included in the report
16	Normal Q-Q plot of residuals	Regression Analysis	SUCCESS	0.10	included in the report
17	Error by target range	Regression Analysis	SUCCESS	0.21	included in the report
18	Error distribution across models	Regression Analysis	SUCCESS	0.11	included in the report
19	Feature importance - Ridge	Feature Analysis	SUCCESS	0.11	included in the report

11. Errors, Warnings and Missing Visualizations

Skipped visualizations (2)

Not applicable to the data or results available - nothing was estimated to fill the gap.

Feature selection: score vs number of features: every configuration used the same number of features

Memory consumption comparison: memory consumption is not measured per configuration by the pipeline; no values are estimated